

Capstone Project Proposal – IS 4110



Project FarmTrust

Group 18

Department of Computing and Information

Systems Faculty of Computing

Sabaragamuwa University of Sri Lanka

October 2025

Approval of Capstone Project

1. Title of the Capstone Project: Project FarmTrust

2. Details of Group members:

Index No.	Name with Initials	Email	Mobile Number	Signature of student
22FIS0515	GUNATUNGA P.Y.U.	22fis0515@ms.sab.ac.lk	0773689599	
22FIS0568	ARIYASENA G.G.S.	22fis0568@ms.sab.ac.lk	0705067302	
22FIS0563	HEWAPATHIRANA L.N.S.	22fis0563@ms.sab.ac.lk	0711983118	
22FIS0567	PERERA J.A.H.G.	22fis0567@ms.sab.ac.lk	0713158065	
22FIS0569	AHNAF M.A.J.A.	22fis0569@ms.sab.ac.lk	0764640282	

3. Name of the Mentor: Mr. W.M.D. Shashimina

4. Mentor's Designation: Senior Software Engineer

5. Mentor's Organization: Cloud Solutions International (Pvt) Ltd

6. Name of the Internal Supervisor: Ms. T. Kayalvizhy

7. Internal Supervisor's Designation: Lecturer (Temp)

For office use only:

Approved / Not approved: -----

Signature of the Supervisor

Signature of the Mentor

Suggestions if any:

Date:

Table of Contents

1.Acknowledgement	4
2.Introduction and Objectives	5
2.1 Introduction	5
2.2 Objectives of the Capstone Project	6
3. Analysis	7
3.1. Feasibility Study	7
3.1.1. Technical Feasibility	7
Assessment: Moderately High Feasibility, with Defined Complexities.	7
3.1.2. Operational Feasibility	8
Assessment: High Feasibility, with user support and progress	8
3.1.3 Economic Feasibility	8
3.1.4 Other Feasibility	9
3.2. ER Diagram	10
4.Requirements	11
4.1. Hardware	11
4.2. Software.....	11
4.2.1. Main Software Requirements.....	11
4.2.2. Technology Stack.....	11
5.Details of Modules.....	12
6.Proposed System	14
6.1 Introduction	14
6.2 Features of the System	14
6.3 Functional Requirements.....	14
6.4 Non-Functional Requirements	15
6.4. Methodology.....	16
7.Gantt Chart	18
8.Cost Analysis	19
9.References.....	20

1.Acknowledgement

We would like to express our deepest gratitude and appreciation to Prof. S. Vasanthapriyan, the Dean of the Faculty of Computing, Sabaragamuwa University of Sri Lanka, Dr. L.S. Lekamge, the Head of the Department of Computing and Information Systems, Faculty of Computing, Sabaragamuwa University of Sri Lanka and Mrs. W.T. Saranga Somaweera Lecturer (Probationary), Capstone Project Coordinator, Department of Computing and Information systems, Faculty of Computing, Sabaragamuwa University of Sri Lanka for their coordination and high supervision of our capstone project. As well as their guidance, support and commitment have been instrumental in ensuring the successful execution of our project.

We are grateful for their invaluable advice, encouragement, and expertise throughout the process. We would also like to extend our gratitude to Ms. T. Kayalvizhy Lecturer (Temp) and Internal Supervisor of the Capstone Project. The mentor of our team, Mr. W.M.D. Shashimina, Senior Software Engineer at Cloud Solutions International (Pvt) Ltd for their contributions and dedication in supervising our team have been pivotal in shaping the direction and progress of our project. We appreciate their valuable insights, technical expertise, and continuous support.

Additionally, we would like to express our thanks to all the faculty members, staff, and fellow students who have provided us with valuable feedback, resources, and assistance during this project. Their input and involvement have been greatly appreciated. Lastly, we thank our friends and family for their unwavering support, understanding, and encouragement throughout this endeavor. Their belief in our abilities has been a constant source of motivation. We are truly grateful for their help and guidance.

2.Introduction and Objectives

2.1 Introduction

FarmTrust is a prototype digital farming system aimed at establishing trust, transparency and intelligence in the farm-to-buyer system in Sri Lanka. By removing all these barriers to farmers, including unfair pricing, payment insecurity, and market access, the platform offers a solution to the long-time standing issues by incorporating AI, blockchain, and financial technology into one inclusive solution.

FarmTrust is not just another online market like the traditional markets, which only connects buyers and farmers. It offers AI-enhanced pricing insights to farmers so that they can make sound decisions and offer fair market prices to their products. An escrow payment system created with the help of a QR/OTP provides secure and transparent exchange, whereas the Trust Mark system implemented with the assistance of the blockchain will list all the transactions that successfully undergo the verification to ensure accountability and credibility among all users.

It also has pre-harvest contracting and bidding to assist farmers in finding buyers in the future and minimizing risks, and offline SMS/USSD gateway to encompass farmers with low internet or smartphone access. Besides, the data analytics dashboard will enable farmers and buyers to gain real-time information to make strategic decisions and enhance productivity.

With FarmTrust, we will establish a sustainable and fair farming market which will empower the farmers, build trust among buyers and help modernize the agricultural sector of Sri Lanka.

2.2 Objectives of the Capstone Project

The main purpose of the FarmTrust project is to create a safe, smart, and open agricultural platform that will eliminate trust gaps between farmers and buyers and will encourage fair trade and financial transparency. There are goals such as:

- **AI-Powered Market Intelligence:** enable farmers to make data driven pricing decisions based on automated smart price reporting analysis based on real time and historical market data.
- **Secure Digital Payments system:** enables fast, secure, and transparent payments between farmers and buyers through verified digital gateways.
- **Cloud Hosting and Scalability** – ensures efficient, secure, and scalable platform performance through cloud infrastructure with automatic scaling, data encryption, and reliable uptime.
- **Pre-Harvest Bidding and Contracting:** Have pre-harvest sales and forward contracts to have market demand and reduce the risk of production.
- **Offline Access through SMS/USSD:** Facilitate the same access to all farmers, even those who are not connected to the internet or do not own a smartphone, by having an easier gateway of SMS/USSD.
- **Data Analytics Dashboard:** Provide actionable information using an interactive analytics dashboard to facilitate improved decision-making efforts on the part of both the farmers and buyers
- **Empowerment and Inclusion:** Go a step further to promote inclusiveness through narrowing down the digital divide and empowering rural dwellers with new Agri-technology.

3. Analysis

3.1. Feasibility Study

A web application that connects the farmers and the buyers through this platform includes features such as AI powered pricing, secure payment methods include online methods, pre-harvest contracting and offline connectivity or availability of access.

3.1.1. Technical Feasibility

Assessment: Moderately High Feasibility, with Defined Complexities.

- **Core Web Platform & Dashboards:**

Building this web application for crop listing, buyers analytically browsing according to the list, and the analytical dashboard is **highly feasible**. These common features can be added in the application using React, Node, Python like technologies and the cloud services like AWS and Google cloud.

- **AI-Powered Price Engine:**

This section is moderately complex compared to others in achieving as a main component to the application. But can be achievable component through few listed steps.

Data access: using the data from the official and the government parties with the collaborative or more convenient feasible; however, the quality and the real-time availability of the data might be challenging. Through the versions of the system, it can be achievable.

Algorithm development: machine learning platform that integrates platform, market, and contextual data to generate the smart pricing report requires specialized data science skills. It is feasible, but it will be an ongoing development task.

- **Payment Gateway:**

Merging with the already built digital payment methods like PayHere, mobile money, and online banking are making it highly feasible to this web application. It is the recommended and most available practice to make security and compliance.

- **Offline-First SMS specification:**

Sending price alerts and notifications via SMS APIs is straightforward.

- **Conclusion:**

The project is technically feasible according to the analysis through a well-skilled team. Initiate with the development through core web platform and SMS notification, then, can lead the system to the AI engine integration

3.1.2. Operational Feasibility

Assessment: High Feasibility, with user support and progress

- **Training and adoption to the system:** The platform is well organized to cover the different user levels. This platform connects buyers and farmers. Especially lower connection facilities where farmers struggle to communicate with the usability of the application like forecasting. Essential for including farmers with basic phones, making it highly accessible.
- **Day-to-Day Management:** The platform automates matching, bidding, and payments.
- **Platform Support:** A customer supported communicable team is mandatory to help the users with transactions, disputes, and technical problems.
- **AI Model:** the essential to this AI pricing engine is continuous monitoring and periodic manual adjustments on the recommendations. Which should be realistic and trustworthy.
- **Pre-Harvest Contract:** The platform must have clear processes for handling scenarios where a crop fails, or a buyer cancels a pre-order, which will require a dedicated support and mediation function.
- **Workflow Integration:** Pre-harvest contracting module seamlessly integrates into the planning cycle. Both farmers and the buyers are adding significant operational values by making riskless.
- **Verdict:** the system is Operationally feasible. The important challenge is all about human based not technical. Providing the best user-based support and making clear fair processes to manage the marketplace's dynamic and the AI recommendations.

3.1.3 Economic Feasibility

There is a high potential room for profit.

- **Cost:**
 - **Development:** high biggining cost for the web and the AI engine.
 - **Infrastructure:** Recurring cloud hosting costs, which will scale with user traffic.
 - **SMS Costs:** Ongoing costs for sending messages, which can be substantial on a scale.
 - **Salaries:** costs for the developers, data engineers, customer assistants, and admin staff.
- **Revenue Streams:**
 - **Transaction Fee:** A small commission on each successful pre-order, bid, or direct sale. This is the primary and most scalable revenue stream.
 - **Subscription Fee:** A premium tier for buyers offering advanced analytics, early access to listings, or a lower transaction fee.
 - **Featured Listings:** from the farmers a small amount of money to list out the availability to the buyers.

- **Return on Investment:**
 - Value is powerful, especially for our nation. By assuring fair prices to both parties and securing the payments, the system can become quickly essential for its users.
 - The business model is designed for high levels once a critical mass of users is achieved and transaction volume grows. The initial phase will require investment to cover the cost before the platform becomes self-sustaining.
- **Verdict:** economically it is feasible with a scalable business model. It is a well match for grant funding. The key to economic success is rapid user acquisition to drive transaction volume and cover the high fixed operational costs.

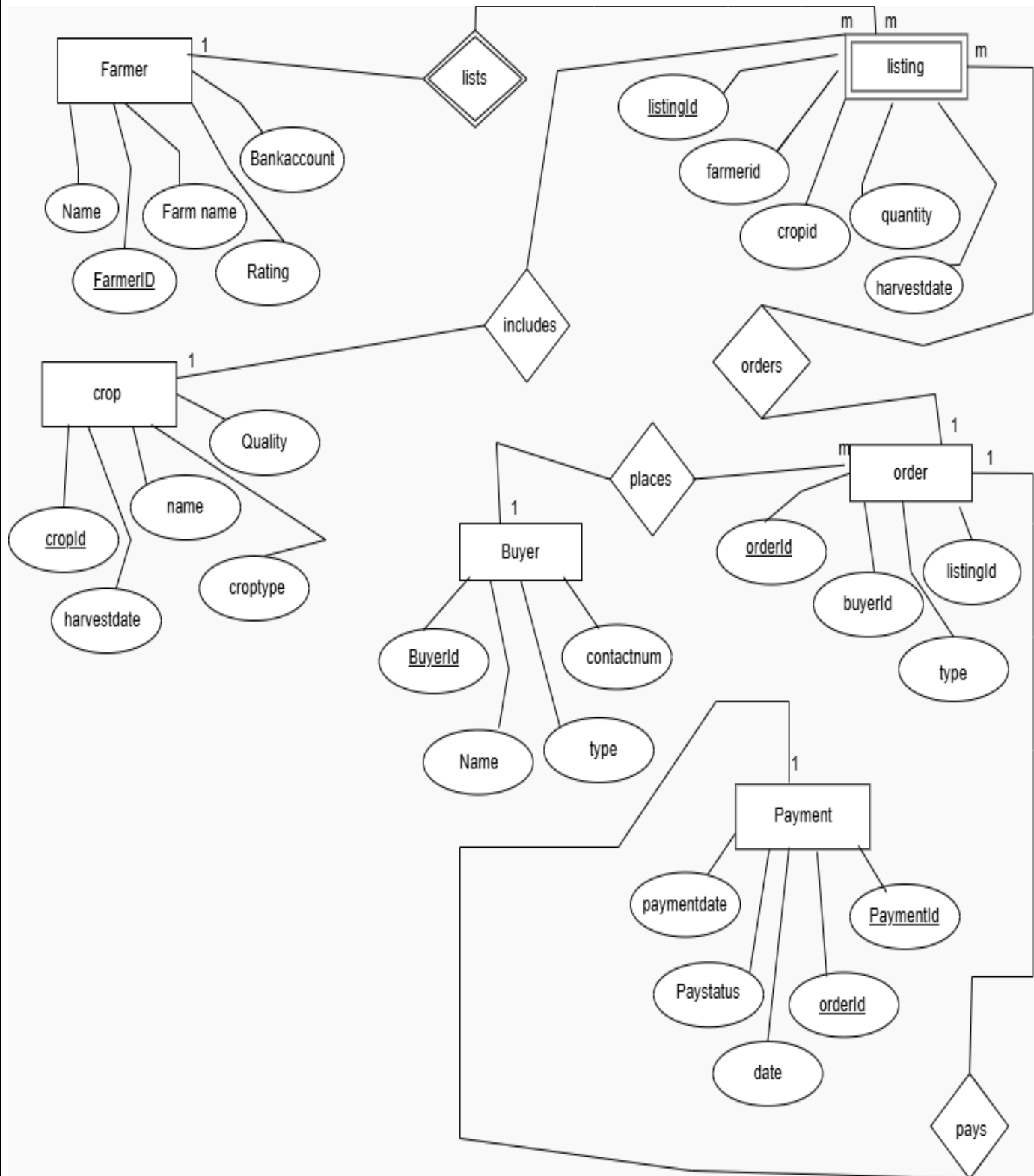
3.1.4 Other Feasibility

- **Legal Feasibility:** The pre-harvest agreements (pre-orders and bids) are legally binding on contracts. The platform's Terms of Service must clearly define the liabilities for both farmers and buyers in case of crop failure or buyer default. The site must comply with all financial rules related to payments. Must follow data protection rules regarding the collection and use of farmer and buyer data. Especially for analytics and AI.
- **Scheduling Feasibility:**
Release the initial version of the platform with crop listing, pre-harvestmen contract, and basic SMS notification in a single pilot region. After that, developing and integrating the AI pricing engine and data analytics dashboards. Attempting to build all features simultaneously would be unmanageable and likely lead to failure.
- **Recommendation:**

The "FarmTrust" platform is a HIGHLY FEASIBLE project with a strong potential for significant impact and profitability.

- **PROCEED:** This project is viable. The recommended strategy is to begin with a detailed project plan and a focused Phase 1 pilot to validate the concept, gather user feedback, and demonstrate traction to investors.

3.2. ER Diagram



4.Requirements

4.1. Hardware

4.1.1. User Devices

- Laptops
- Mobile Devices
- 8 GB RAM
- 64-bit Operating Systemx64 – based processor
- Intel(R) Core (TM) i5-6200U CPU 2.3GHz Processor
- Storage – 500GB HDD Or SSD
- Internet Connection
- Mouse/Keyboard

4.2. Software

4.2.1. Main Software Requirements

- Django / Node.js
- Figma
- Visual Studio Code, PyCharm
- GitHub
- MySQL /MongoDB
- TensorFlow, Pandas, NumPy
- Postman, Selenium
- AWS / Google Cloud / Microsoft Azure

4.2.2. Technology Stack

- Diagram Creator
- Figma
- Android Studio
- Flutter
- Visual Studio
- GitHub
- Django

5.Details of Modules

- ❖ Farmer Module - It is applied to ensure that the farmers have complete control over the listing of their products and payment history.
 - Farmer Registration and profile management.
 - Listing of crops including type, quantity, estimated date of harvest, price, and photos.
 - Accept or reject bids from buyers.
 - Track order, payment, and delivery status.
- ❖ Buyer Module - This module assists the buyers to associate with credible farmers and buy quality produce in an open market.
 - Registration and verification of buyers.
 - Search and narrow down crops available in the region, type, or certification.
 - Place pre-orders or bids.
 - Rate farmers and provide feedback after successful transactions.
- ❖ Admin Module - The admin will make sure that all activities in the marketplace are equitable, safe, and well regulated.
 - Controlling user accounts (Buyers and Farmers).
 - Confirm the authenticity of the farmers and their crops.
 - Create reports and control system analytics.
- ❖ Payment Module - It also provides payment operations within the platform that are smooth, reliable, and traceable.
 - Safe online payment through mobile wallets or banks.
 - Confirms successful transactions and updates farmer payment records.
 - Generate transaction receipts for both customers and farmers. and keeps simple records of transactions to be referred to.
- ❖ Product Listing and Verification Module - This module is used to guarantee the transparency of the products and the assurance of buyers.
 - Allow farmers to add, update, or delete crop listings.
 - Showing valid and authorized products to the customers.
 - Controlling the product image, description, and availability.
 - Authenticating and filtering listings to only those which are valid, displayed on the platform.

- ❖ Analytics and Recommendation Module - It will help farmers to plan production and optimize predictability according to data analytics.
 - Market pricing and farmer performance insights.
 - Suggestions on crops that are profitable or buyers.
 - Visualization of supply demand trends.

- ❖ Crop Tracking Module - It ensures transparency and authenticity throughout the process of delivery.
 - Tracking crop movement and delivery status in real time.

Sending pickup, delivery, and verification update notifications.

6. Proposed System

6.1 Introduction

The suggested FarmTrust system is a web-based platform that can be hosted on the cloud and is created to establish a direct connection between the farmers and the buyers without involving the middlemen who tend to dictate the prices in the agricultural markets. The platform guarantees transparency, equitable payments, and safe transactions as well as giving farmers an advantage to access the market better and buyers a verified and trustworthy source.

The system combines a Trust Mark reputation system, a Secure Digital Payment Gateway, and an AI-based Pricing Assistant to make sure the prices are fair and reliable. It is stored in Microsoft Azure Cloud which guarantees scalability, constant availability and safety of data to all users.

6.2 Features of the System

- User Management – Provides separate registration and login facilities for farmers, buyers, and administrators with role-based access.
- Crop Listing and Search – Farmers can list crops with details; buyers can browse, search, and filter listings easily.
- Secure Digital Payment System – Integrated with PayHere, mobile money, and online banking for secure payments and instant notifications.
- Trust Mark Reputation System – Each farmer receives a trust mark score based on transactions and feedback to ensure transparency.
- AI-Powered Pricing Assistant – Analyzes market trends to suggest fair prices for crops before listing.
- Cloud Hosting and Scalability – Hosted on Microsoft Azure Cloud for automatic scaling, high performance, and data backups.

6.3 Functional Requirements

Administrator

- Verify and approve farmer and buyer registrations.
- Manage and monitor user activities, transactions, and feedback.
- Handle complaints, generate reports, and ensure compliance with policies.

Farmer

- Register securely and create verified profiles.
- Add, update, or remove crop listings with relevant details.
- View orders, confirm transactions, and receive payments directly.
- Track sales analytics and trust mark score.

Buyer:

- Register and manage accounts with secure login.
- Browse and filter crop listings, view details, and place orders.
- Make secure payments and receive notifications.
- Rate farmers and leave reviews after purchases.

Trust Mark System

- Calculates a score for each farmer based on transactions and feedback.
- Automatically updates after every successful sale.
- Display scores publicly to promote credibility.

Secure Digital Payment System

- Handles payments via PayHere and online banking with encryption.
- Stores transaction history for transparency and dispute resolution.

AI-Powered Pricing Assistant:

- Suggests optimal price ranges for crops based on historical and market data.
- Improve decision-making for farmers with price insights.

Cloud Hosting and Scalability:

- Ensures continuous uptime through Azure with auto-scaling and backups.

6.4 Non-Functional Requirements

- Performance – The system should process requests within 2–3 seconds under standard load.
- Scalability – Cloud infrastructure must scale automatically with demand.
- Security – SSL/TLS encryption, hashed passwords, and PCI-compliant payment gateways.
- Reliability – Azure Cloud ensures 99% uptime with redundancy and backups.
- Usability – Responsive interface for desktop and mobile browsers.
- Maintainability – Modular architecture allows easy updates and future expansions.
- Availability – 24/7 accessibility nationwide through Azure hosting.
- Data Integrity – Validated data entries to prevent duplication or loss.
- Compliance – Adheres to Sri Lankan data protection and e-commerce laws.

6.4. Methodology

The development of FarmTrust will follow an **Agile Methodology** prioritizing iterative development, flexibility and close cooperation with farmers and buyers. It would mean ensuring the system meets users' needs and allows for changes along the way.

1. Requirement Gathering and Analysis

- i. Preliminary Meetings and field visits: Take the time to arrange for stakeholder meetings to better comprehend the challenges as well as to capture the spirit of market and farm site visits.
- ii. User Stories and Use Cases: Outline functional and non-functional requirements for user stories. These requirements will guide the development of each feature.
- iii. Feature Prioritization: Work with stakeholders to establish the order of development. The trust core features and inclusion features will be prioritized in the first phase of development.

2. System Design

- i. Architecture Design: It should be designed from a Cloud Infrastructure-focused perspective on high availability, elastic scalability and security for FarmTrust. Besides that, it would articulate the system design, data flow, and the integration of AI, payment integration, and SMS/USSD interfaces.
- ii. Database Design: Database schema should design relationships and develop an ER diagram between the entities, Farmer, Buyer, Crop Listing, Transaction, Bid/Contract, Trust Mark, and Smart Pricing Data.
- iii. UI/UX Design: Develop wireframes and prototypes with a focus on intuitive and user-centered design for the Farmer App/SMS interface and the Buyer Web/Mobile App. Design iteration will occur after usability testing, specifically for the SMS/USSD Gateway and Integrated Data Analytics Dashboard.

3. Development

- i. Agile Sprints: Break down the system into mini sprints, each focusing on the key functionalities like Crop Listing, AI Pricing, and Bidding. Hold frequent reviews to ensure stakeholder satisfaction.
- ii. Continuous Integration: Implement CI/CD pipelines to automate system testing and seamless code updates at regular intervals. Deployments without difficulties

4. Implementation of Key Features

- i. AI-Powered Pricing & Market Intelligence: Use AI to study the market and platform data to compile a Smart Pricing Report and deliver tailored recommendations for the farmers.
- ii. Secure Digital Payment System: Implement secure gateways (e.g. *Payhere*, mobile money) for direct payments. Make instant payments, notifications, and tracking available to users.
- iii. Offline-First SMS/USSD Gateway: Allow farmers to list crops, check prices and confirm deliveries through SMS/USSD with no internet access.
- iv. Pre-Harvest Contracting & Bidding Module: Let farmers list crops early, accept bids or pre-orders and give buyers visibility into future supplies.
- v. Integrated Data Analytics Dashboard: Turn transaction and market data into insights for farmers and buyers and provide intuitive dashboards.

6. Deployment

- i. Staging Environment: The system will be deployed in a cloud-based development setting for final performance functionality testing.
- ii. Production Deployment: Activate the authorized cloud-hosted FarmTrust system for every user.

7. Maintenance and Updates

- i. Post-launch Support: Provide ongoing support to solving problems and track AI Pricing progress.
- ii. Regular Updates: Implement required enhancements and add new features based on performance data and user feedback.

8. Documentation

- i. Technical Documentation: Provide complete documentation for developers and maintainers on the software, cloud configuration, system design, and APIs.
- ii. User Manual: Allow farmers and buyers instructions and tutorials describing how to use Analytical Dashboard and SMS/USSD interface.
- iii. Reports & Records: Keep records of all transactional logs, payments, audit records and information to ensure transparency, enable challenges and confirm analytics data.

7.Gantt Chart

Activity	Time Duration (Weekly)													
	First Month				Second Month				Third Month				Fourth Month	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Project Planning														
Requirements Gathering and Analysis														
Learning Techniques														
Design														
Front-end Development														
Back-end development														
Documentation														
Testing														
Deployment and Maintenance														

8. Cost Analysis

Cost Item	Description	Approximate Value (LKR)
Development Cost		
Frontend Development	Building Responsive web Interfaces using ReactJS and Tailwind CSS	-
Backend Development	Spring Boot+ REST API+ authentication + payments	-
Database	Storing user, crop, and transaction data	-
Hosting and Deployment	Deploying and hosting the web application	-
Payment Gateway	Online payments between farmers and buyers	-
AI Integration	Crop price prediction model	1000
Testing Costs		
Testing and QA tools	Browser compatibility and API testing	-
Emails and SMS Notifications	Sending system notifications	-
Maintenance Costs		
Server Maintenance	Cloud hosting and server upkeep	1000
Total Cost		~2000

9.References

[1] “Market Information,” *Krushilanka.lk*, 2025. <https://krushilanka.lk/index.php/en/market-information-en> (accessed Oct. 27, 2025).